

Advanced Machine Learning

DATA 642

Lecture 11 — Practical Considerations and Closing Remarks

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

Selecting the number of clusters

We cannot use the inertia or the silhouette score because they both assume that the clusters are spherical. Instead, we can try to find the model that minimizes a theoretical information criterion such as the Bayesian Information Criterion (BIC) or the Akaike Information Criterion (AIC):

$$BIC = \log(N)p - 2 \log(\hat{L})$$

$$AIC = 2p - 2 \log(\hat{L})$$

- N is the number of instances.
- p is the number of parameters learned by the model.
- $\hat{L}$ is the maximized value of the likelihood function of the model. This is the conditional probability of the observed data $\mathbf{X}$, given the model and its optimized parameters.

Both BIC and AIC penalize models that have more parameters to learn (e.g., more clusters), and reward models that fit the data well (i.e., models that give a high likelihood to the observed data).

Other algorithms for anomaly detection

1. **Fast-MCD (Minimum Covariance Determinant):**

- Useful for outlier detection and dataset cleanup.
- Assumes normal instances are generated from a single Gaussian distribution, but dataset is contaminated with outliers.
- Estimates parameters of Gaussian distribution while ignoring potential outliers, resulting in better estimation of elliptic envelope and improved outlier identification.

2. **Isolation Forest:**

- Efficient algorithm for outlier detection, especially in high-dimensional datasets.
- Builds a Random Forest where each Decision Tree randomly selects features and thresholds to isolate instances.
- Anomalies, being far from other instances, tend to get isolated in fewer steps than normal instances.

3. **Local Outlier Factor (LOF):**

- Compares instance density to density around its neighbors.
- Anomalies are often more isolated than their nearest neighbors.

4. **One-Class SVM (Support Vector Machine):**

- Suited for novelty detection.
- Attempts to separate instances in high-dimensional space from the origin.
- Finds a small region encompassing all instances; new instance falling outside this region is considered an anomaly.
- Hyperparameters include those typical for SVMs, plus a margin hyperparameter for the probability of a new instance being mistakenly considered novel.
- Works well with high-dimensional datasets but does not scale to large datasets like other SVMs.